Agriculture: Corn Pest Management Guidelines

Corn Stunt

Spiroplasma kunkelii

Symptoms and Signs

Corn stunt, as the name implies, results in stunted plants(/PMG/S/D-CN-SKUN-FS.001.html). Severely infected plants and those infected early in their development may be only 5 feet tall with very short internodes, rather than the usual 10-12 feet in height. The stalk may have multiple ears, sometimes as many as 6-7 on a single plant. The ears are small and do not fill properly leaving a large number of blank spaces on the ear. The kernels that do develop are frequently "loose" leading to what is called "loose tooth ears(/PMG/S/D-CN-SKUN-GH.001.html) ". Younger leaves near the top of the plant are yellow. With age they take on a reddish to reddish purple color that varies by variety.

Comments on the Causal Agent

Spiroplasma kunkelii is a bacterial-like organism known as a spiroplasma. Corn leafhoppers, *Dalbulus maidis,* carry the spiroplasma from diseased corn to healthy corn. The spiroplasma overwinters within the adult leafhopper; when the leafhoppers emerge from overwintering in early spring, they can be infective. Disease symptoms appear about 3 weeks after the corn is infected. The disease is most severe in corn planted after July 1 but can occur in corn planted as early as March and April.

Comments on the Disease

All current commercial varieties of field and sweet corn appear to be susceptible. Yield loss depends on the growth stage of the corn when it is infected but can be significant, especially in late corn. Until recently, the occurrence of both the leafhopper and corn stunt were sporadic in the southern San Joaquin Valley. Since 1996, however, the leafhopper and corn stunt disease have occurred on a yearly basis with significant losses each year.

Management

Early planting is the best way to minimize damage from this disease. There is no chemical treatment to control the spiroplasma, and chemical treatment for the controlling the leafhoppers is generally not effective in controlling the disease.

UC PEER REVIEWED

UC IPM Pest Management Guidelines: Corn
UC ANR Publication 3443

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Text Updated: 01/06